



Using a Venn diagram to display outcomes for two events

Task A: Venn diagrams

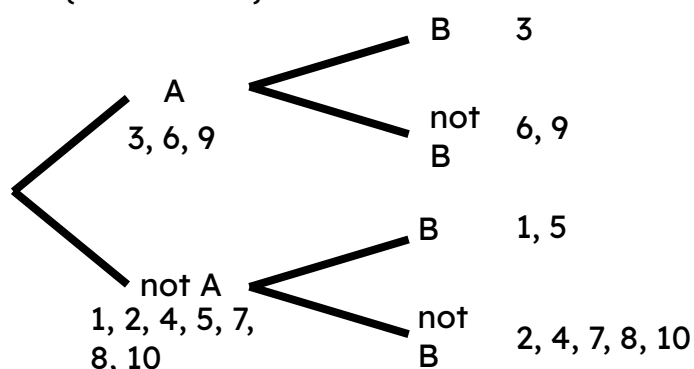
1) Here is an outcome tree for two events.

Draw a **Venn diagram** to show the same information.

The two events are:

$A = \{\text{multiple of } 3\}$

$B = \{\text{factor of } 15\}$



2) Here is an outcome table for two events.

Draw a **Venn diagram** to show the same information.

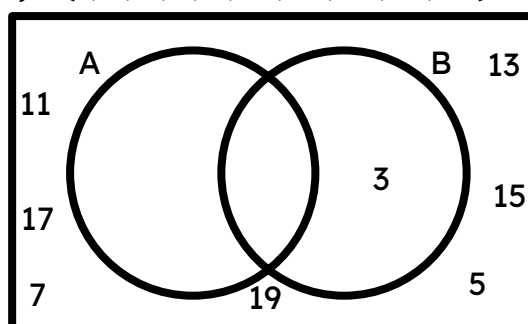
	prime number (P)	not a prime (P')
factor of 30 (F)	2 3 5	1 6 10 15
not a factor of 30 (F')	7 11 13	4 8 9 12 14

3) Here is a partially complete outcome table and **Venn diagram**.

Complete both diagrams.

	<u> </u> (A)	not a square number (A')
factor of 18 (B)	1 9	
not a factor of 18 (B')		5 7 11 13 15 17 19

$\xi = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$



Task B: Outcomes from Venn diagrams

1) Use this **Venn diagram** to answer the following questions:

a) List the outcomes that are in event A.

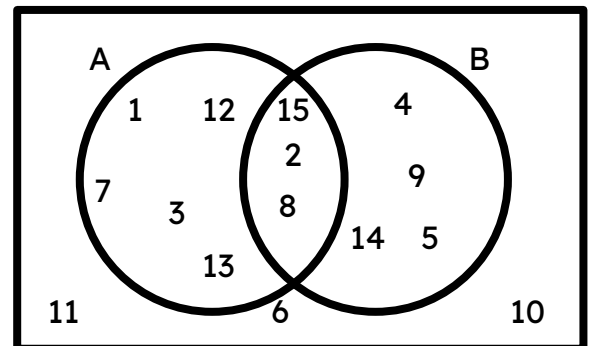
$\xi = \{\text{integer values from 1 to 15 inclusive}\}$

b) List the outcomes that are in event B.

c) List the outcomes that are in both event A and event B.

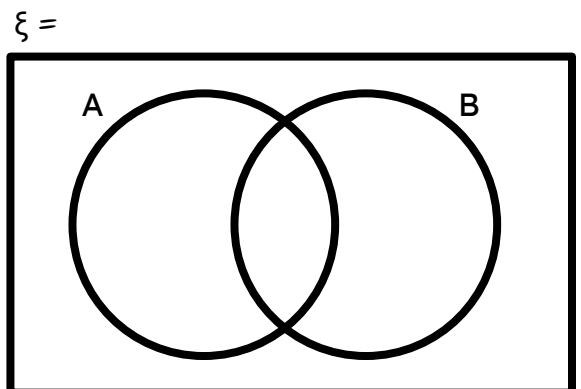
d) List the outcomes that are in event A but not in event B.

e) List the outcomes that are not in either event A nor event B.



2) a) Use the following information to complete the **Venn diagram**.

- the possible outcomes from the trial are the integers from 1 to 10 inclusive
- there are five possible outcomes for event A
- $\{6\}$ is the only outcome of both events
- $\{2, 5, 8\}$ are not outcomes of event A nor event B
- the outcomes that are only in event B are prime



b) List the outcomes of event A.

c) List all the outcomes for event A or event B or both.

3) Use this **Venn diagram** to answer the following questions:

a) How many outcomes are in the sample space?

b) How many outcomes are in event A?

c) How many outcomes are in event B?

d) How many outcomes are not in event A or event B?

e) Explain why your answers to parts b, c and d do not sum to part a.

